Apple Log 2

White Paper
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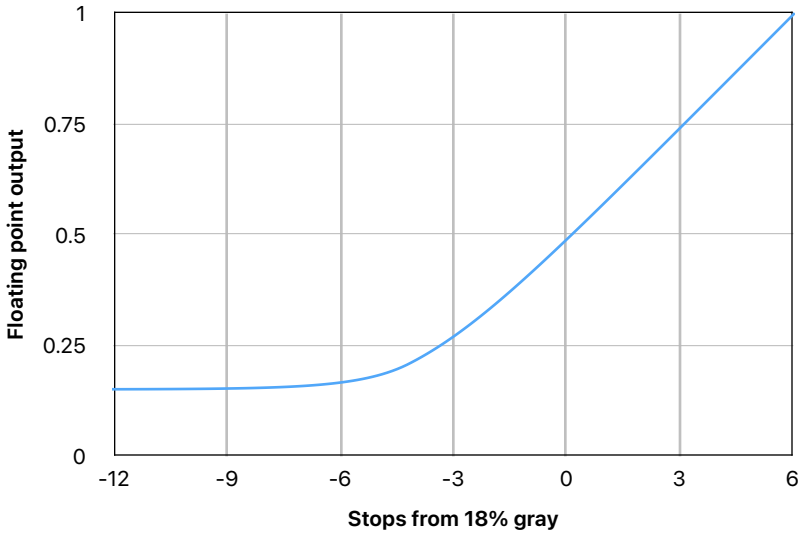
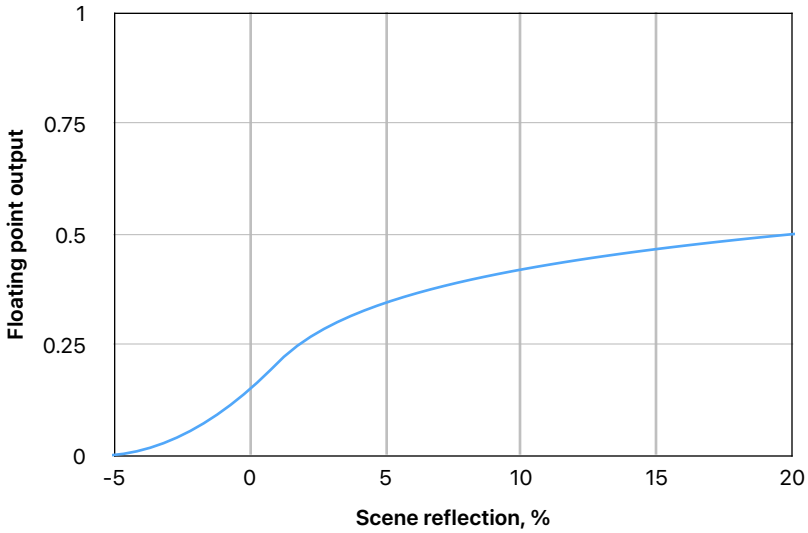
Introduction

The Apple Log 2 color space is a scene-referred encoding using a logarithmic curve and a wide color gamut. It is designed to best preserve video data at capture time, with minimal processing. Log captures are not intended for direct viewing; the output rendering is applied by an additional display transform. Apple Log 2 was developed to provide the flexibility required by production and post-production workflows, such as live monitoring, visual effects, and color grading. These captures are characterized by a transfer function, and a set chromaticities, yielding a color space optimized for the color sensors found in iPhone.

Transfer function

The transfer function converts between the signal proportional to the scene reflectance and the pixel values saved in the movie file. The Apple Log 2 color space uses the Apple Log transfer function, which has been optimized for sensors with pixel sizes typically found in mobile devices such as iPhone. The function is a logarithmic curve smoothly transitioning into a parabola in order to better capture negative signal values for improved preservation of the noise signal component.





Scene reflection	Encoded signal	
	Float	10 bit, full range
0%	0.150477	154
18%	0.488272	500
90%	0.681686	697
1200%	1.0	1023

Encoding function

The encoding function is implemented inside the camera and is provided for informational purposes only. This function converts linear scene reflection values to the output data captured by the camera. The conversion is given by the equation

$$\mathrm{P} = f(\mathrm{R}) = \begin{cases} \gamma \log_2(\mathrm{R} + \beta) + \delta, & \text{if } \mathrm{R} \geq \mathrm{R}_t \\ c(\mathrm{R} - \mathrm{R}_0)^2, & \text{if } \mathrm{R}_0 \leq \mathrm{R} < \mathrm{R}_t \\ 0, & \text{if } \mathrm{R} < \mathrm{R}_0 \end{cases}$$

with

$$\begin{aligned} \mathrm{R}_0 &= -0.05641088 \\ \mathrm{R}_t &= 0.01 \\ c &= 47.28711236 \\ \beta &= 0.00964052 \\ \gamma &= 0.08550479 \\ \delta &= 0.69336945 \end{aligned}$$

The scene reflection signal R captured by the camera is represented using a floating point encoding. The R value of 0.18 corresponds to the signal produced by an 18% reflectance, reference gray chart.

Decoding function

The decoding function can be applied to the captured footage in order to recover the linear scene reflection signal. The decoding function is given by the equation

$$\mathrm{R} = f^{-1}(\mathrm{P}) = \begin{cases} 2^{\frac{\mathrm{P} - \delta}{\gamma}} - \beta, & \text{if } \mathrm{P} \geq \mathrm{P}_t \\ \sqrt{\frac{\mathrm{P}}{c}} + \mathrm{R}_0, & \text{if } 0 \leq \mathrm{P} < \mathrm{P}_t \\ \mathrm{R}_0, & \text{if } \mathrm{P} < 0 \end{cases}$$

where

$$\mathrm{P}_t = c(\mathrm{R}_t - \mathrm{R}_0)^2$$

The captured pixel value P is using a floating point encoding normalized to the $[0,1]$ range.

Apple Wide Gamut

Apple Wide Gamut describes how colors are represented by the Apple Log 2 color space. Apple Log 2 uses RGB primaries, and its associated color gamut has been optimized for the use of Apple camera sensors in scene-referred workflows. The white point is set to the D65 illuminant.

	CIE chromaticity coordinates	
	x	y
Red primary (R)	0.725	0.301
Green primary (G)	0.221	0.814
Blue primary (B)	0.068	-0.076
White point (D65)	0.3127	0.3290

RGB / $Y'C'_BC'_R$ Conversion

Where applicable, such as in an Apple ProRes-encoded QuickTime file, Apple Log 2 color space uses the conversion between RGB and luminance-chrominance $Y'C'_BC'_R$ representation as defined in the ITU-R BT.2020-2 specification. This conversion is governed by the following formula:

$$Y' = 0.2627R' + 0.6780G' + 0.0593B'$$

$$C'_B = \frac{B' - Y'}{1.8814}$$

$$C'_R = \frac{R' - Y'}{1.4746}$$

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